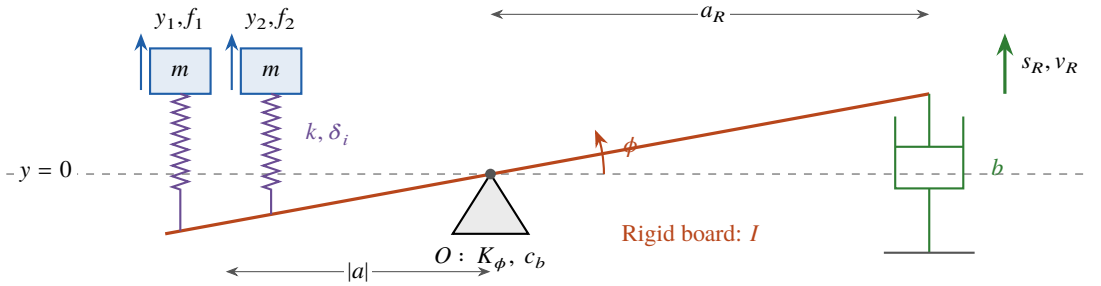
y_1, f_1
y_2, f_2
m
m
k, δ_i
a_R
s_R, v_R
$y = 0$
ϕ
b
Rigid board: I
O : K_ϕ, c_b
$|a|$